

Problem 1) Scalars in computing are simply a container for an item.
The item itself may have internal structure.
Mathematical scalars are like physical scalars, but without units.

Problem 2) No. $F = m\vec{a} = \text{mass}$

Problem 3) $\varphi = 0$

Problem 4) $\varphi = \pi$

Problem 5) $\varphi = \pi/2$

Problem 7) $\vec{e} \cdot \vec{e} = \frac{\vec{a}}{|\vec{a}|} \cdot \frac{\vec{a}}{|\vec{a}|} = \frac{|\vec{a}| |\vec{a}|}{|\vec{a}| |\vec{a}|} = 1$

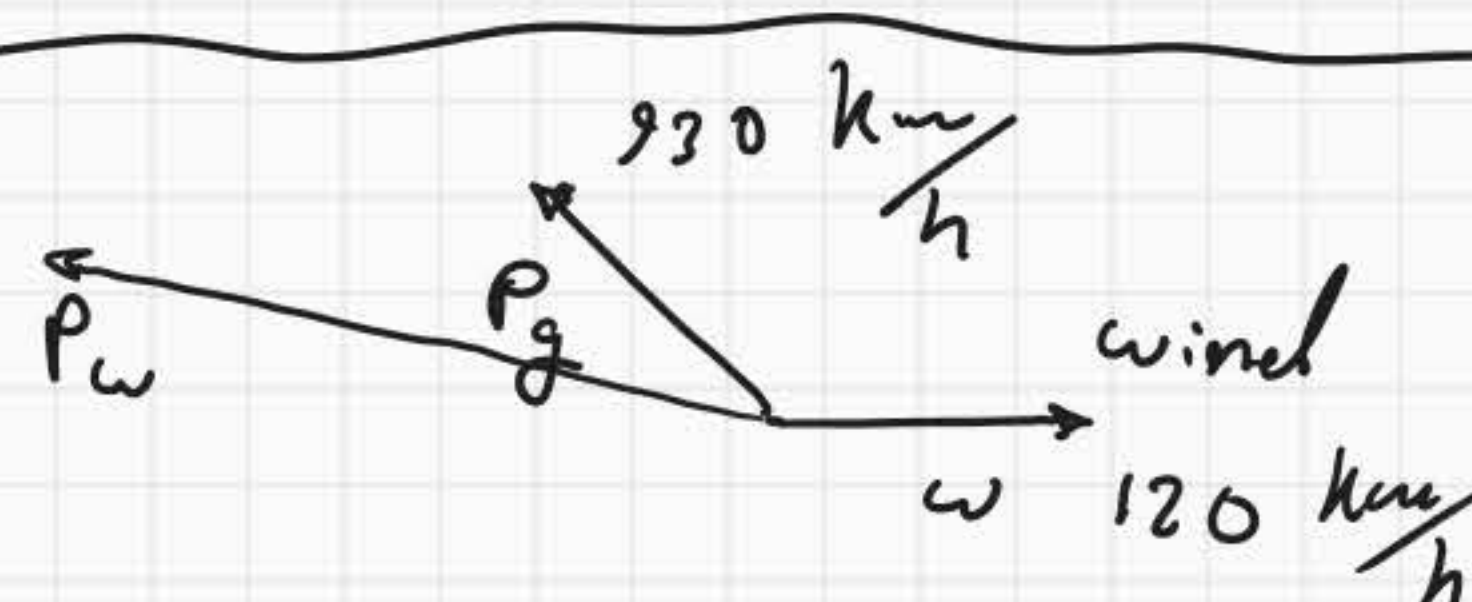
Problem 8)
$$\begin{aligned} \vec{a} \cdot \vec{b} &= a_1 b_1 (\vec{e}_1 \cdot \vec{e}_1) + a_1 b_2 (\vec{e}_1 \cdot \vec{e}_2) + a_1 b_3 (\vec{e}_1 \cdot \vec{e}_3) \\ &\quad + a_2 b_1 (\vec{e}_2 \cdot \vec{e}_1) + a_2 b_2 (\vec{e}_2 \cdot \vec{e}_2) + a_2 b_3 (\vec{e}_2 \cdot \vec{e}_3) \\ &\quad + a_3 b_1 (\vec{e}_3 \cdot \vec{e}_1) + a_3 b_2 (\vec{e}_3 \cdot \vec{e}_2) + a_3 b_3 (\vec{e}_3 \cdot \vec{e}_3) \\ &= a_1 b_1 + a_2 b_2 + a_3 b_3 \end{aligned}$$

Problem 9) $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$; denote $b_i + c_i = d_i$

$$\begin{aligned} \vec{a} \cdot (\vec{b} + \vec{c}) &= a_1 d_1 + a_2 d_2 + a_3 d_3 = a_1 b_1 + a_1 c_1 + a_2 b_2 + a_2 c_2 + \\ &\quad a_3 b_3 + a_3 c_3 \\ &= \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} \end{aligned}$$

Problem 10) $\vec{P}_w = ?$
$$= \vec{P}_g - \vec{w}$$

The rest follows immediately.



Problem 11) $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \varphi \Rightarrow \varphi = \cos^{-1} \left(\frac{a_1 b_1 + a_2 b_2 + a_3 b_3}{|\vec{a}| |\vec{b}|} \right)$

Problem 12) when $\theta = 0$

Problem 13) $\hat{e}_3 = \hat{e}_1 \times \hat{e}_2 \quad \hat{e}_2 = \hat{e}_3 \times \hat{e}_1 \quad \hat{e}_1 = \hat{e}_2 \times \hat{e}_3$

Problem 14) $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$

$$\vec{a} \equiv (a_x, a_y, a_z) \quad ; \quad \vec{b} \equiv (b_x, b_y, b_z) \quad ; \quad \vec{c} \equiv (c_x, c_y, c_z)$$

$$\begin{aligned} \vec{a} \times (\vec{b} + \vec{c}) &= (a_x \hat{i} + a_y \hat{j} + a_z \hat{k}) \times ((b_x + c_x) \hat{i} + (b_y + c_y) \hat{j} + (b_z + c_z) \hat{k}) \\ &= (\cdot) \hat{i} \times \hat{i} + (\cdot) \hat{j} \times \hat{j} + (\cdot) \hat{k} \times \hat{k} \\ &\quad + a_x (b_y + c_y) \hat{i} \times \hat{j} + a_x (b_z + c_z) \hat{i} \times \hat{k} \\ &\quad + a_y (b_x + c_x) \hat{j} \times \hat{i} + a_y (b_z + c_z) \hat{j} \times \hat{k} \\ &\quad + a_z (b_x + c_x) \hat{k} \times \hat{i} + a_z (b_y + c_y) \hat{k} \times \hat{j} \\ &= a_x (b_y + c_y) \hat{k} + a_x (b_z + c_z) \hat{j} \\ &\quad - a_y (b_x + c_x) \hat{k} + a_y (b_z + c_z) \hat{i} \\ &\quad - a_z (b_x + c_x) \hat{j} - a_z (b_y + c_y) \hat{i} \\ &= (a_y b_z - a_z b_y) \hat{i} + (a_x b_z - a_z b_x) \hat{j} + (a_x b_y - a_y b_x) \hat{k} \\ &\quad + (a_y c_z - a_z c_y) \hat{i} + (a_x c_z - a_z c_x) \hat{j} + (a_x c_y - a_y c_x) \hat{k} \\ &= (\vec{a} \times \vec{b}) + (\vec{a} \times \vec{c}) \quad \text{QED.} \end{aligned}$$

Problem 15) The area of the base of the parallelepiped can be expressed as $|\vec{b} \times \vec{c}|$. Then, the volume of $\sim$ will be

$$V = h |\vec{b} \times \vec{c}| = \vec{a} \cdot (\vec{b} \times \vec{c}) = \vec{b} \cdot (\vec{c} \times \vec{a}) = \vec{c} \cdot (\vec{a} \times \vec{b})$$

Problem 16) a)
$$\begin{cases} (1, a, b) \cdot (4, 3, 0) = 0 \\ (1, a, b) \cdot (5, 1, 7) = 0 \end{cases} \Rightarrow \begin{cases} 4 + 3a + 0 = 0 \\ 5 + a + 7b = 0 \end{cases} \Rightarrow a = -\frac{4}{3}$$

$$\Rightarrow b = \frac{1}{7} \left(\frac{4}{3} - 5 \right) = -\frac{11}{21}$$

b)
$$\begin{aligned} \vec{a} &\equiv (1, 7, 0) \\ \vec{b} &\equiv (1, 1, 1) \end{aligned} \quad \left\{ \begin{aligned} \vec{a} \times \vec{b} &= (a_y b_z - a_z b_y, a_x b_z - a_z b_x, a_x b_y - a_y b_x) \\ &= (7, 1, -6) \end{aligned} \right.$$

c)
$$\begin{aligned} |\vec{a} \times \vec{b}| &= |\vec{a}| |\vec{b}| \sin \theta \\ \vec{a} \cdot \vec{b} &= |\vec{a}| |\vec{b}| \cos \theta \end{aligned} \Rightarrow \begin{aligned} |\vec{a} \times \vec{b}|^2 &= |\vec{a}|^2 |\vec{b}|^2 \sin^2 \theta \\ (\vec{a} \cdot \vec{b})^2 &= |\vec{a}|^2 |\vec{b}|^2 \cos^2 \theta \end{aligned}$$

$$\Rightarrow |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2 |\vec{b}|^2 (1 - \cos^2 \theta) = (|\vec{a}| |\vec{b}| \sin \theta)^2 = |\vec{a} \times \vec{b}|^2$$

Problem 17) Prove $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})$

$$\vec{a} = a_x \hat{x} + a_y \hat{y} + a_z \hat{z}$$

$$\vec{b} = b_x \hat{x} + b_y \hat{y} + b_z \hat{z}$$

$$\vec{c} = c_x \hat{x} + c_y \hat{y} + c_z \hat{z}$$

$$\begin{aligned} \Rightarrow \vec{a} \times (\vec{b} \times \vec{c}) &= \left[a_y (b_x c_y - b_y c_x) - a_z (b_z c_x - b_x c_z) \right] \hat{x} \\ &\quad + \left[a_z (b_y c_z - b_z c_y) - a_x (b_x c_y - b_y c_x) \right] \hat{y} \\ &\quad + \left[a_x (b_z c_x - b_x c_z) - a_y (b_y c_z - b_z c_y) \right] \hat{z} \end{aligned}$$

Meanwhile, the RHS is,

$$\begin{aligned} \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b}) &= (a_y c_y b_x + a_z c_z b_x - a_y b_y c_x - a_z b_z c_x) \hat{x} \\ &\quad + (a_x c_x b_y + a_z c_z b_y - a_x b_x c_y - a_z b_z c_y) \hat{y} \\ &\quad + (a_x c_x b_z + a_y c_y b_z - a_x b_x c_z - a_y b_y c_z) \hat{z} \end{aligned}$$

Comparing the two expressions, it follows that;

$$\boxed{\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b})} \quad \checkmark$$

Problem 18) Prove $(\vec{a} \times \vec{b}) \times \vec{c} = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{a}(\vec{b} \cdot \vec{c})$

Recall from lecture 2, the anti-commutativity property of the cross product: $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$

$$\Rightarrow (\vec{a} \times \vec{b}) \times \vec{c} = -\vec{c} \times (\vec{a} \times \vec{b})$$

From the BAC-CAB rule proved above, it then follows that

$$\boxed{(\vec{a} \times \vec{b}) \times \vec{c} = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{a}(\vec{b} \cdot \vec{c})} \quad \checkmark$$